

Ollscoil
Teicneolaíochta
an Atlantaigh

Atlantic
Technological
University

Data Science

> Selecting & merging data

Dropping observations

- We can drop unwanted attributes very easily.
- We can look at the headers of the data frame using the **names()** function.
- Then we can use this information to select the data we want into a new data frame.

- Eg only requiring Q3 and Q4 data

- **Exercise...**

- Use names extract Age and Q1

Deleting attributes from data

```
# This command shows all attributes where Q3 or Q4 are contained
```

```
# The ! operator reverses this choice
```

```
include_list <- names(managers) %in% c("Q3", "Q4")
```

```
include_list
```

```
# This list can be used to extract this data
```

```
new_data <- managers[(include_list)]
```

```
new_data
```

Creating a subset

- How can we perform a comparison between date ranges?
- We can use the **subset()** function

```
# Using the subset function  
# to extract all records from my_data where age > 35 or age < 24. Only select the  
# listed attributes
```

```
attach(managers)  
new_data <- subset(managers, Age >= 35 | Age < 24, select = c(Q1, Q2, Q3, Q4))  
new_data
```

How would we select a subset of managers called new_managers where gender = M and age > 25. Show all attributes between Gender and Q4 only.

Creating a random sample

- Sampling from larger datasets is common practice in data mining and machine learning.
- We may want to select two random samples, creating a predictive model from one and testing from the other.
- The **sample()** function enables us to take a random sample (with or without replacement) of size n from a dataset.
- We can take a random sample of size 3 from our **managers** dataset using the following statement:

Selecting a random sample from my_data my_sample <-

managers[sample(1:nrow(managers), 3, replace = FALSE),]
my_sample



Start

End No of
elements

Replace
elements?

Merging datasets horizontally

- To merge two data frames (datasets) horizontally, you use the **merge()** function.
- In some cases, two data frames are joined by one or more common key variables (that is, an inner join).
- For example, **total <- merge(dataframeA, dataframeB, by="ID")**
 - ⑥ merges dataframeA and dataframeB by ID.
- Similarly **total <- merge(dataframeA, dataframeB, by=c("ID","Country"))**
 - ⑥ merges the two data frames by ID and Country.
- Horizontal joins like this are typically used to add variables to a data frame.

Merging datasets horizontally

- Alternatively we can join two data frames horizontally without the need to specify a common key **total <- cbind(A, B)**
- This function horizontally concatenates objects A and B.
- For the function to work properly, each object must have the same number of rows and be sorted in the same order.

Merging datasets vertically

- Use **rbind()** to add datasets vertically.
- **total <- rbind(dataframeA, dataframeB)**
- The two data frames must have the same variables, but they don't have to be in the same order.
- If dataframeA has variables that dataframeB doesn't, then before joining them, do one of the following:
 - Delete the extra variables in dataframeA.
 - **OR** create the additional variables in dataframeB, and set them to **NA** (missing).
- Vertical concatenation is typically used to add observations to a data frame.

Apply these techniques by merging a new dataset called “MoreData” with the managers dataset